

COURSE TITLE : COMPUTER AIDED PRESS TOOL DRAWING LAB
SEMESTER : 6
COURSE CODE : 6117
COURSE CATEGORY : A
PERIODS/ WEEK : 7
PERIODS/ SEMESTER : 105
CREDITS : 6

TIME SCHEDULE

MODULE	TOPIC	PERIODS
1	Introduction to Pro-E Sketching Basic modeling Advanced modeling	25
2	2D & 3D Drawing (By Auto Cad)Assembling Introduction to surface Modeling	28
3	Introduction to mould design	25
4	Compound tool (By Solid works) Bending tool, Forming tool, Drawing tool, Deep drawing tool	27
	TOTAL	105

COURSE OUT COMES

On completion of the course, Student should be able to:

1. Understand computer aided press tool drawing by Solid works, Pro-E Software's
2. Understand assembling the components.
- 3.Appreciate the mould design and drawing.
4. Understand to draw compound tools by solid works.

SPECIFIC OUTCOMES

1.1.0 Understand computer aided press tool drawing by Solid works, Pro-E Software's

- 1.1.1 Draw a sketch using tool bars-dimensioning-2D drawings
- 1.1.2 Identify Basic modeling-protrusion-Extrude
- 1.1.3 Create advanced modeling

2.1.0 Understand assembling the components

2.1.1 Draw -projection view--Detailed view-Dimensioning

2.1.2 Describe introduction to surface modeling-conversion of surface to solids

3.1.0 Analyze mould design and drawing

3.1.1 Study introduction to mould design

3.1.2 Draw compound tool-Bending tool-Forming tool-Drawing tool

4.1.0 Understand to draw compound tools by solid works.

4.1.1 Understand to draw a bending tool.

4.1.2 Understand to draw a forming tool.

4.1.3 Understand to draw a drawing tool.

CONTENT OUTLINES.

MODULE-I

Introduction to Pro-E-Familiarization of interface/Windows.

Sketching- Drawing a sketch using sketcher tool bars – dimensioning – constraints – importing 2D drawings.

Basic modeling-Protrusion-Extrude-Single & double sided, Mirror. Rib-Chamfer, Fillet& Shell.

Holes - Standard & cut – Revolve.

Creating datum planes, axes, curves , points, co – ordinate systems.

Sweep – Blend – Patterns – straight, rotational & general.

Advanced modeling – Variable section sweep, swept blend, helical sweep etc.

Assembling – Assembling the components – exploded state.

Drawing by auto cad- General view – projection view – detailed view – auxiliary view – Revolved section view – dimensioning with geometric tolerances.

Introduction with to surface modeling – Creating surfaces – open and close ends – merge tools – trim – intersection – conversion of surface to solids.

Introduction to mould design – creating components – add shrinkage – parting line – split core and cavity lay out for multy impression cooling – runners – gates – water line – Add ejector pins – BOM.

Compound tool – Direct and indirect knockout on compound tool – Bending tool – V bending tool, U bending tool , L bending tool , multiple angle bending tool.

Forming tool- Solid type and pressure pad type.

Drawing tool – shallow and deep drawing tool.

Note: First three modules should be completed using Auto cad & Pro-E.

Design and Assembly drawings of press tools module – 4by using Solid works.